HRITIZ GOGOI

RESEARCH INTERESTS

Discrete Optimization, Operations Research, Game Theory.

EDUCATIONAL BACKGROUND

PhD candidate *June 2022 - present*
Industrial Engineering and Operations Research (IEOR)
Indian Institute of Technology Bombay, Mumbai
Supervisors: Prof. Vishnu Narayanan and Prof. Avinash Bhardwaj
Expected date of completion: September, 2027

NSR Pre-Doctoral Fellow *June 2021 - May 2022*
Decision Sciences
Indian Institute of Management, Bangalore
Mentor: Prof. Ishwar Murthy

Master of Science(MSc) Economics *April 2017 - 2019*
Indian Institute of Technology Roorkee

Bachelor of Technology(BTech) Civil Engineering *2011-2015*
National Institute of Technology Silchar

RESEARCH

PhD research:

- *On Exactness of SDP relaxation for the Maximum Cut Problem*, co-authored with Avinash Bhardwaj, Vishnu Narayanan and Abhishek Pathapati:
  - Characterizes graph classes with exact SDP relaxations for Max-Cut, establishes uniqueness conditions and rank-preserving operations, and resolves two open problems of Mirka and David Williamson (2024).
  - Submitted to *Discrete Optimization*, arXiv preprint uploaded with application.
- *Max-k-Cut via Node Features*, co-authored with Avinash Bhardwaj and Vishnu Narayanan.
  - Studies SDP exactness for Max-k-Cut via node features; introduces a greedy vector-balancing algorithm with worst-case guarantees and a computable optimality certificate.
  - Submitted to *Operations Research Letters*; arXiv preprint uploaded with application.

Other research:

- *Lagged Effects of Rainfall on Malaria: A Case Study of Meghalaya*, co-authored with Holendro Singh Chungkham and Strong P. Marbaniang,
  - Uncovers interaction effects between lagged rainfall levels on malaria incidence in Meghalaya state using a distributed lag nonlinear model.
  - Published in *Indian Journal of Public Health*, 69(1):p 57-62, Jan-Mar 2025. Article uploaded with application.

TEACHING ASSISTANTSHIP

- IE 617 Linear Systems 2025 Spring. *Instructor: Prof. Mallikarjuna Rao.*

- IE 209 Linear Optimization and Network Flows 2025 Autumn, 2026 Autumn (ongoing), *Instructor: Prof. Vishnu Narayanan.*
- IE 617 Linear Systems 2024 Spring. *Instructor: Prof. Madhu Belur and Prof. Narayan Rangarajan.*
- IE 609 Mathematical Optimization Techniques 2023 Autumn, 2024 Autumn. *Instructor: Prof. Vishnu Narayanan.*
- IE 684 IEOR Lab 2023 Spring. *Instructor: Prof. Manjesh Hanawal.*

TEACHING INTERESTS

- Operations Research / Optimization Techniques (Linear, Integer, and Nonlinear Programming).
- Production and Operations Analysis.
- Managerial/Industrial Economics.
- Quantitative Methods for Management.

OTHER PROJECTS

IE Department, Purdue University, Indiana, USA *May- August 2024*
Summer Internship Project(*Supervisor: Prof. Seokcheon Lee*)

- Orbital Vehicle Routing Problem for space debris removal.

IEOR, Indian Institute of Technology Bombay, Mumbai *Jan-April 2023*
IE716: Integer Programming Course Project(*Instructor: Prof. Ashutosh Mahajan*)

- Project Name: Single Truck Pickup and Delivery Problem.
- Link: `https://tarunbisht.com/projects/0011-single-vehicle-pickup-and-delivery/`

PROFESSIONAL ENGAGEMENT

Indian Statistical Institute North-East Centre, Tezpur, Assam *August 2019 - March 2020*
Project linked person
Project Supervisor - Prof. Holendro Singh Chungkham

PHD RESEARCH PRESENTATION

- Minisymposium talk. *June 2026*
  Title: *On Exactness of Semidefinite Programming Relaxation for the Maximum Cut Problem*
  SIAM Conference on Optimization (OP26), University of Edinburgh, United Kingdom.
  Link: `https://meetings.siam.org/sess/dsp_programsess.cfm?SESSIONCODE=87896`.
- Poster presentation. *May 2025*
  Geometry, Algorithms and Probablity Workshop. ICTS, TIFR, Bengaluru.
  Link: `https://www.icts.res.in/event/page/31891`

OTHER WORKSHOPS ATTENDED

- Operations Research and Data Analytics in Transportation and Logistics (ODTL 2026) at IIT Bombay. *October 2025*
- Large Scale Optimization (LSO) Summer School at IIT Bombay. *June 2025*
- Mixed Integer Programming (MIP) International Workshop at IIT Bombay. *December 2024*

- Poster presentation on topic "*Irrationality in Financial Markets*" in TAPMI-MAX Planc Winter School on Bounded Rationality at TAPMI, Manipal, Karnataka. *Jan 2018*

VOLUNTEERING WORK

- IEOR Department representative *July 2024- July 2025*
  IRSCP (Institute Research Scholar Companion Program)
- Volunteer, Mixed Integer Programming International Workshop 2024 *December, 2024*
  IIT Bombay
- Volunteer, IMRDC CONFERENCE 2022 *January, 2022*
  IIM Bangalore (Online Conference)
- Routine volunteer for PhD/MTech admission process for IEOR IIT Bombay.

COMPETITIVE EXAMS

- Ranked in National Top 5% in GATE Mathematics (Graduate Aptitute Test for Engineering) 2022
- Ranked in National Top 5% in IIT-JAM MA (Joint Admission Test for MSc- Mathematics) 2017
- Ranked in National Top 5% in AIEEE (All India Examination for Engineering Entrance) 2011

PERSONAL INFORMATION

- Orcid ID: 0009-0005-0216-1232
- LinkedIn ID: `https://www.linkedin.com/in/hritiz-gogoi-48999435a/`
- Emails: hgogoi@iitb.ac.in, hritiz.gogoi@gmail.com.